

Advanced Light Intensity Indicator

EE3024 - Digital Signal Processing

Group: 04

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1. Introduction and System Overview

The ALII (Advanced Light Intensity Indicator) is a hardware device that use a LDR (Light Dependent Resistor) to monitor the surrounding light, filter the signal, and to show the intensity on a seven segment display between 0 and 7. It offers two more features, a long term average intensity assessment and an output stability system to prevent responds to rapid variations in light.

The front end includes a 2nd order Sallen-key low-pass filter, which was selected after careful calculation of the values. This filter has a consistent cutoff behavior and strong reduction of the 50-100Hz noise present in the LDR's supply, to guarantee that only clear and significant light data entered the digital section. The intensity is then mapped to the 0-7 scale by processing the filtered voltage.

Experiments were done with several methods during the design phase to implement the average and stability features which contained lengthy intervals ranging from 30-300s and 300-900s. Initial attempts for analog averaging using integrator circuit was later changed to the digital domain to use flip-flops, registers, and gating logic which proved to be the most reliable and manageable method after testing in the simulators.

Overall, the final system consists of,

- i. 2nd order Sallen-key Low-pass analog filter
- ii. LDR sensing and seven segment level display
- iii. Digital stability logic
- iv. Digital average intensity calculation with switch

This architecture use only gate integrated circuits (ICs), counters, flip-flops, and analog components to make a smooth, accurate, and consistent light-level indication.

2. Complete circuit diagram



3. Feature by feature implementation

a. Analog filtering Stage

The purpose of the analog filtering stage is to remove electrical noise present which can cause flickering or unstable readings in artificial lighting environments. An analog low-pass filter was created and tested to guarantee a clear and dependable input signal.

In the first stage, the topology was decided after calculating the filter order based on the following,

Passband: 15Hz

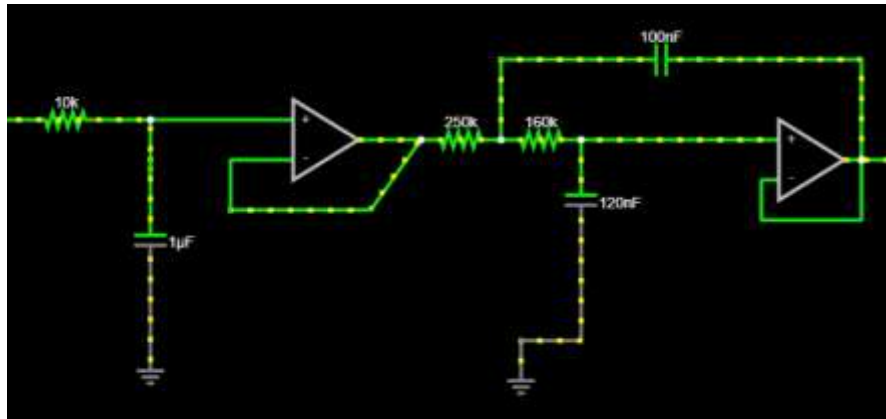
Stopband: 100Hz

Attenuation: 99%

CHANGE ACCORDING TO NEW FILTER

$$\begin{aligned} 20 \log_{10} \left(\frac{1}{100} \right) &= A = -40 \text{ dB} \\ 10 \log_{10} \left(1 + \epsilon^2 \left(\frac{\omega}{\omega_c} \right)^{2n} \right) &= 40 \\ \epsilon^2 &\approx 1 \\ 10^4 &= 1 + \left(\frac{\omega}{\omega_c} \right)^{2n} \\ 10^4 - 1 &= \left(\frac{\omega}{\omega_c} \right)^{2n} \\ \omega &= 100 \text{ Hz}, \omega_c = 15 \text{ Hz} \\ n &= \frac{1}{2} \log_{\left(\frac{100}{15} \right)} (10^4 - 1) = 2.427 \approx 3 \end{aligned}$$

Based on this we implemented a 3rd order filter by cascading a 2nd order Sallen-key low pass with 1st order RC low-pass filter



To match better with the requirements which needed no energy in the 50-100Hz range, the stopband was reduced to 50Hz. Since having a filter with very high orders will reduce the performance in the real life application, the attenuation was reduced to a value which wasn't going to effect the output a lot.

Based on the following,

Passband: 15Hz

Stopband: 50Hz

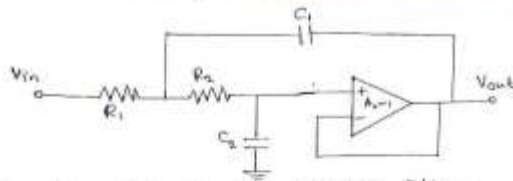
Attenuation: 90%

CHANGE ACCORDING TO NEW FILTER

$$\begin{aligned}
 20 \log_{10} \left(\frac{10}{100} \right) &= A = -20 \text{ dB} \\
 10 \log_{10} \left(1 + \epsilon^2 \left(\frac{\omega}{\omega_c} \right)^{2n} \right) &= 20 \\
 \epsilon^2 &\approx 1 \\
 10^2 &= 1 + \left(\frac{\omega}{\omega_c} \right)^{2n} \\
 10^2 - 1 &= \left(\frac{\omega}{\omega_c} \right)^{2n} \\
 \omega &= 50 \text{ Hz}, \omega_c = 15 \text{ Hz} \\
 n &= \frac{1}{2} \log_{\left(\frac{50}{15} \right)} (10^2 - 1) = 1.908 \approx 2
 \end{aligned}$$

Based on this we implemented a 2nd order filter using a Sallen-key low pass. This required lesser components than the previous version and the op-amp based active filter made sure the filtering was stable and clean than using a passive filter.

CHANGE ACCORDING TO NEW FILTER



Sallen Key Second order lowpass filter.

$$f_c = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}}$$

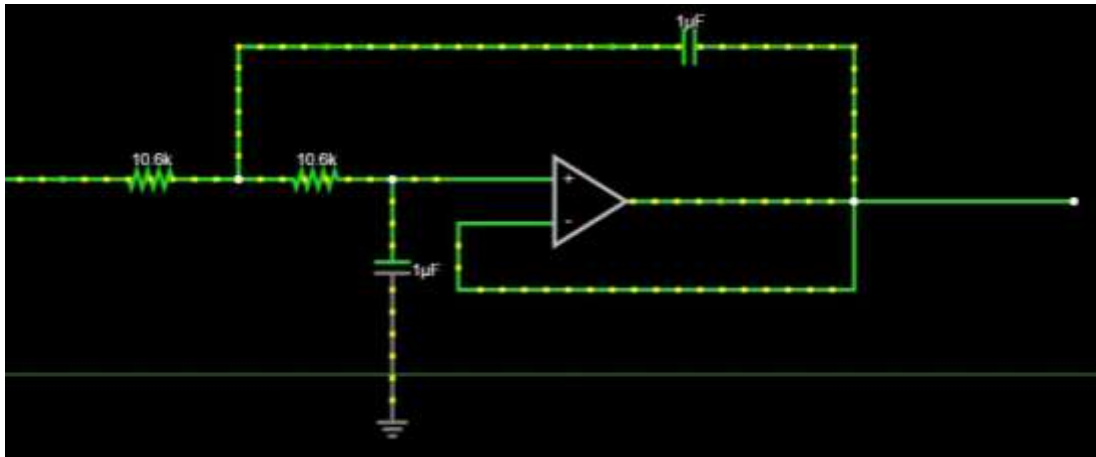
let $C_1 = C_2 = C$ and $R_1 = R_2 = R$

Then $f_c = \frac{1}{2\pi R C}$

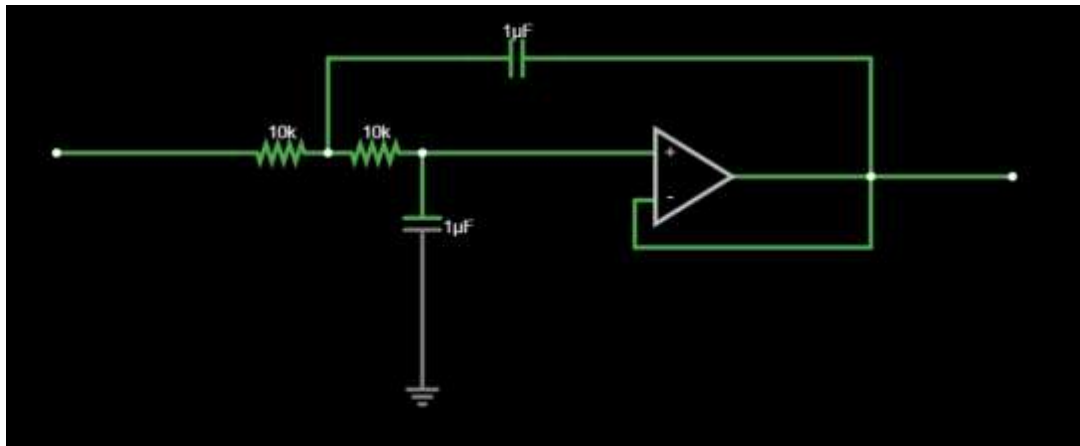
we will choose $C = 1\mu F$.

$$R = \frac{1}{2\pi C f_c}$$

$$R = \frac{1}{(2\pi \times 10^6 \times 15)} = 10.81 \text{ k}\Omega$$



After checking the simulation, the resistor values were changed to real world obtainable values and the circuit was finalized to,



b. LDR Sensing and Seven segment display

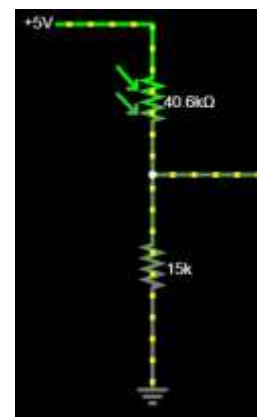
This stage is to convert the light falling on the LDR to a measurable voltage.

The LDR changes its resistance depending on the intensity of light. For **bright light**, lower resistance and **low light** higher resistance. The LDR is positioned in a voltage-divider setup to transform this resistance into a voltage using a suitable resistor value.

Voltage divider formula:

$$V_{LDR} = V_{cc} \cdot \frac{R_{LDR}}{R_{LDR} + R_{fixed}}$$

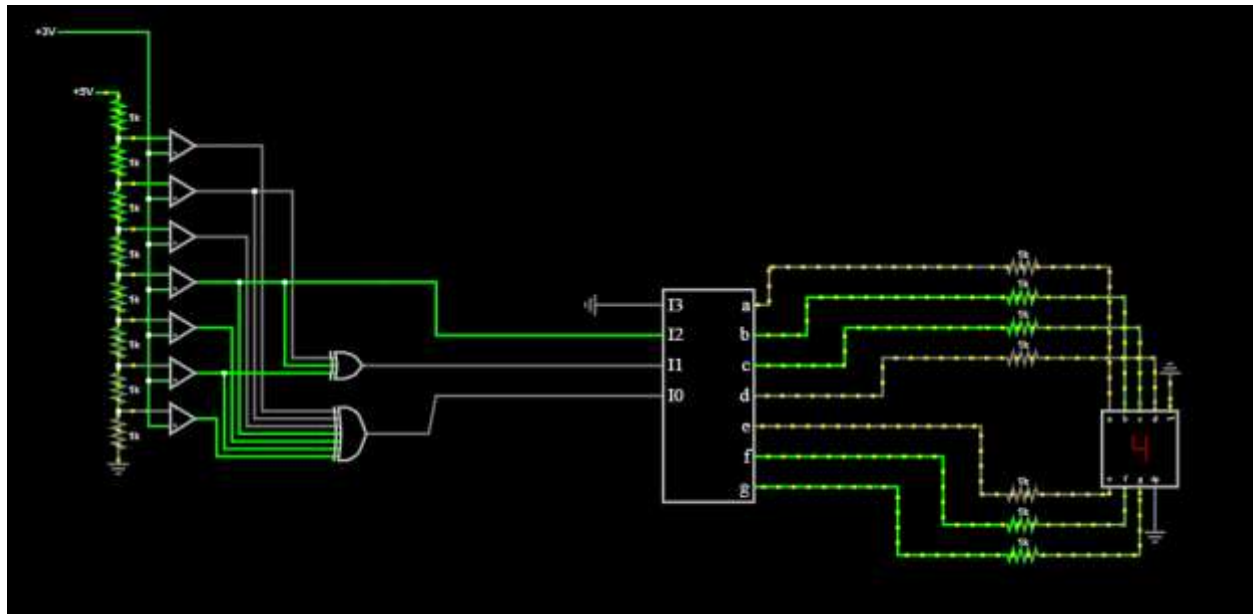
This gives a voltage that increase or decrease with the LDR light.



The analog voltage to digital signal conversion is done using a comparator circuit which divides the voltage into 8 levels from 0-7 and using an 8-3 priority encoder it is converted to BCD. The BCD is connected to the 7 segment display which works as follows

0 = lowest intensity

7 = highest intensity



(//IMAGE OF COMPARATORS 8-3 AND 7SD)

c. Stability feature

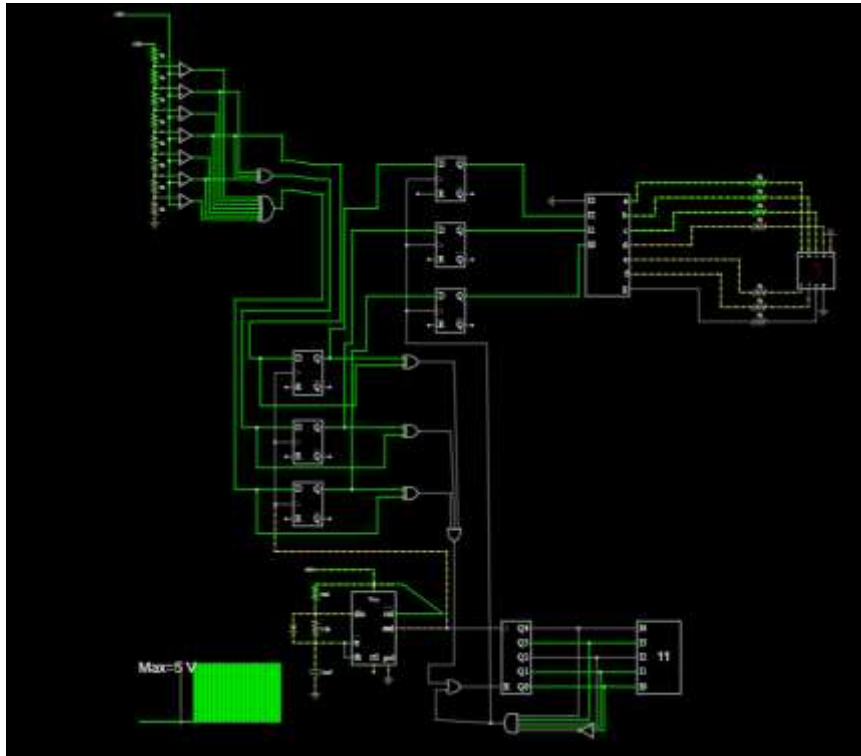
The stability feature makes sure that only after the surrounding light has been constant for continuous duration of 30-300s does the system update the displayed light level. To achieve this, a moving average has been taken in that time range. This stops transient light variations, like shadows passing by, to give unstable results.

Initial design was a combination of analog integrator and binary counter

(//IMAGE OF INTEGRATOR STABILITY)

Later it was found out that the integrator stage wanted an amplifier gain of 1 to have ideal behavior. But in practice only 100k gain amplifiers were available. This made our design invalid. After multiple attempts of trying to do this feature in the analog domain, it was switched to the digital domain.

Using flip-flops, gate logic(adders), the binary counter, and a switch to turn this feature on/off when needed the stability feature was implemented.



//IMAGE OF DIGITAL STABILITY

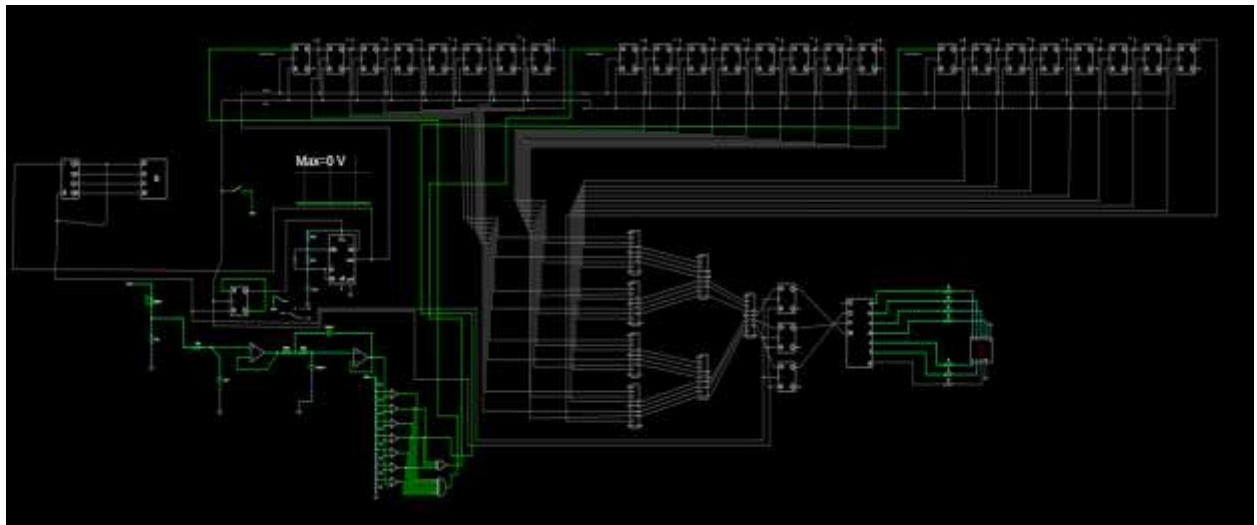
d. Average light intensity module

The average light intensity feature is designed to give a long term perspective of lighting conditions. This feature determines the mean light intensity over a time window of 300-900 seconds and the display shows the average light level from 0-7. Because of this feature, applications like energy management and daylight monitoring can use the ALII.

The digital domain was used to implement the averaging feature, the sampling clock was binary, accumulator registers were used to keep the information for the time period, and the adders would get the mean of the sample by adding them and diving it by the number of samples. Then a BCD to 7-segment was used to display the results in the 7-segment display.

This feature was implemented to get the average when a push button in pressed.

//IMAGE OF DIGITAL AVERAGE



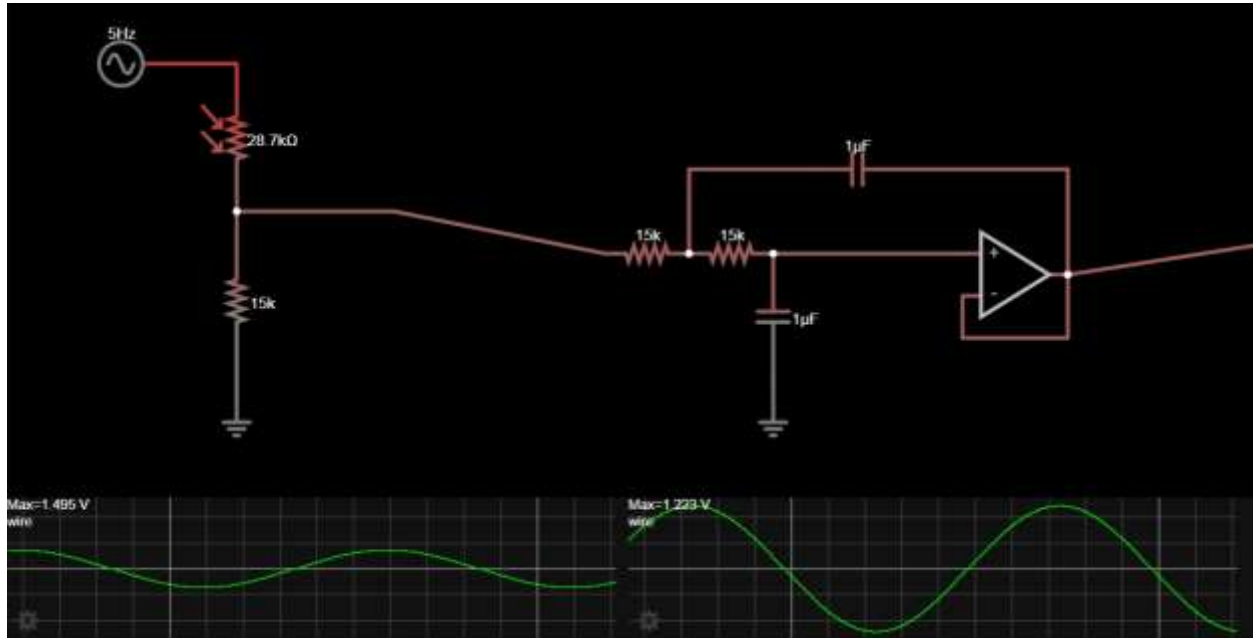
4. Simulation Results

This section includes the key simulation outputs obtained by testing each major subsystem of the ALII module.

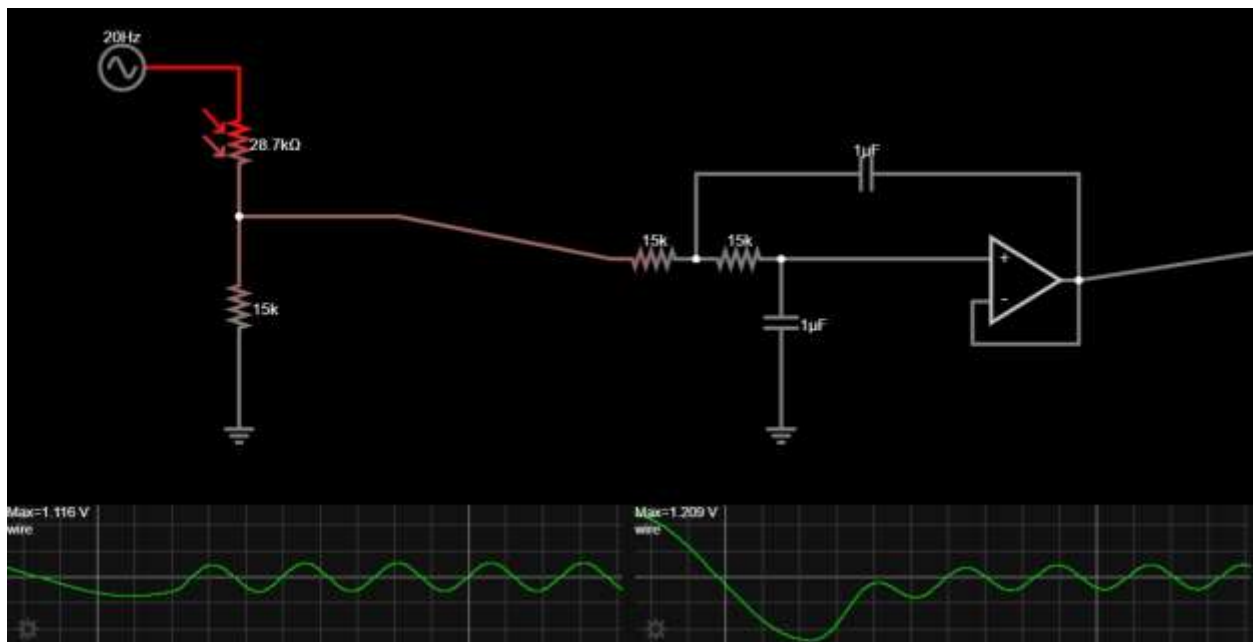
a. Analog Filtering Stage

//IMAGE OF LOW PASS FILTER OUTPUT IN FREQUENCIES

At 5Hz.

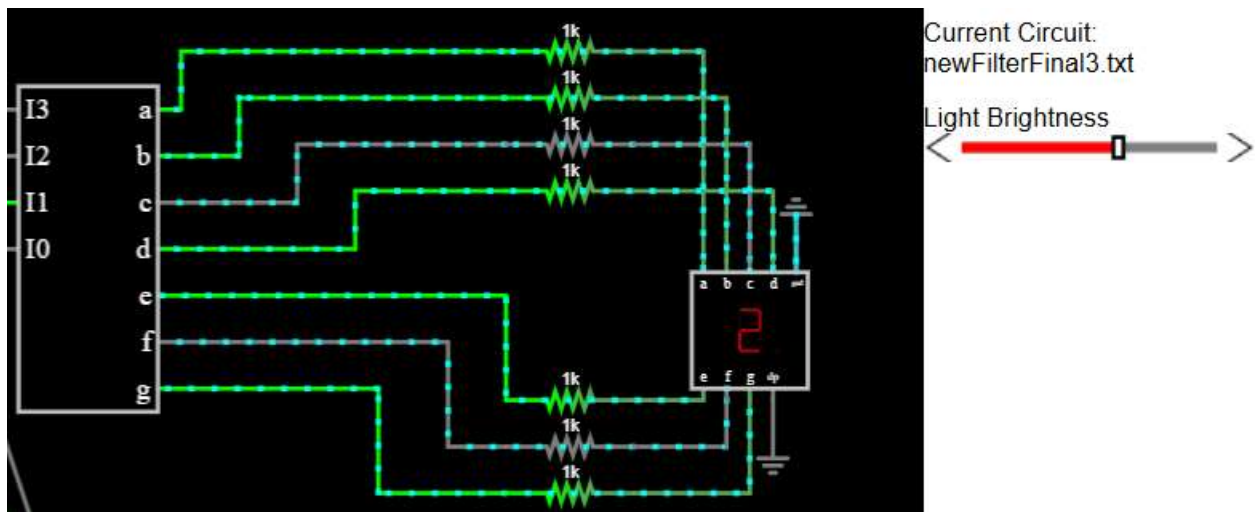
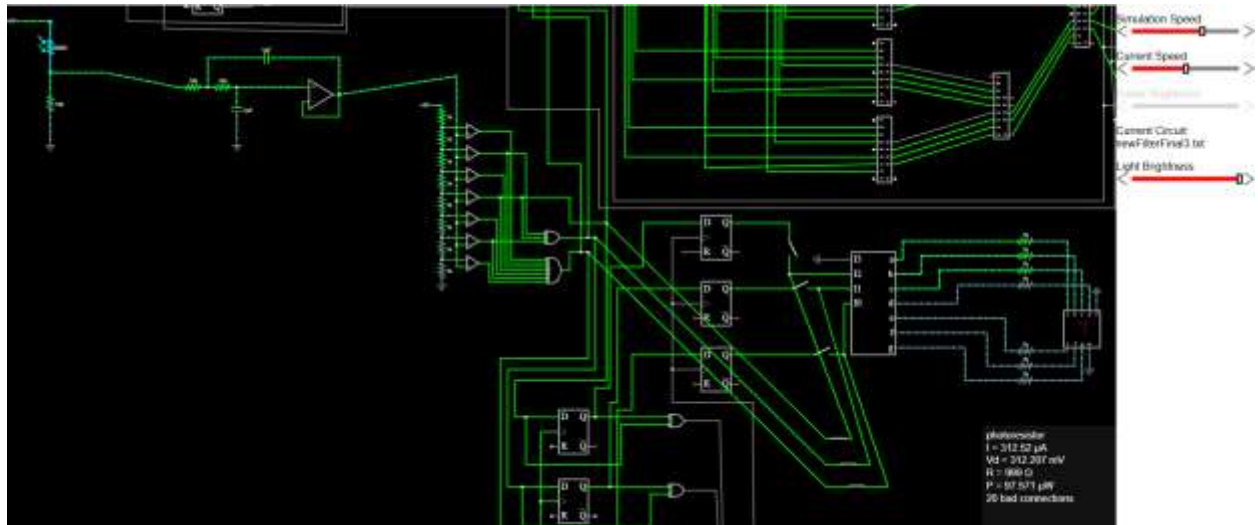


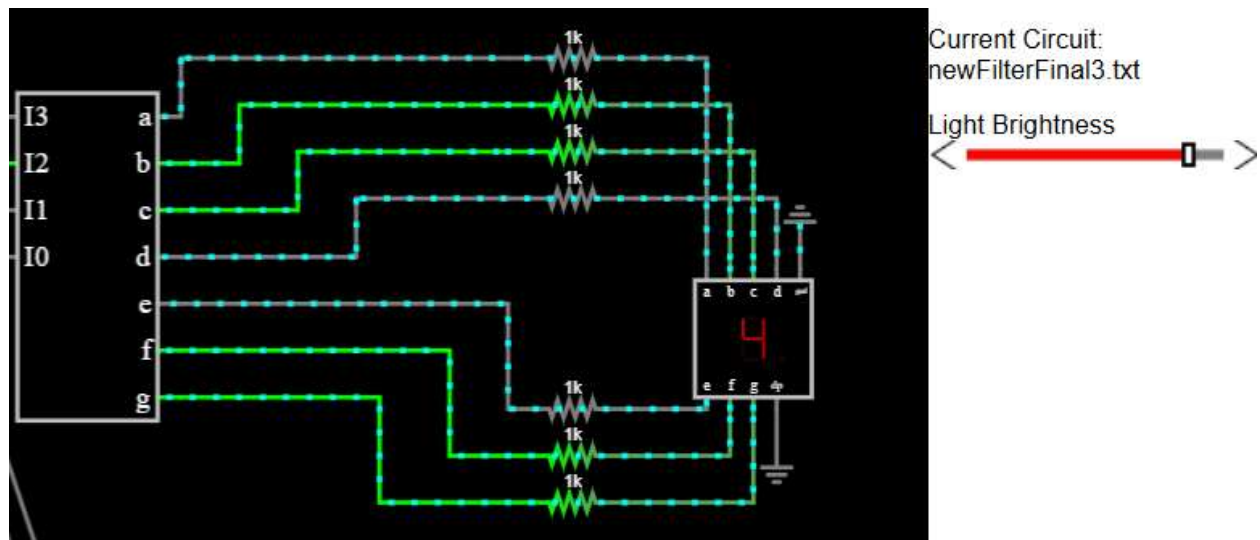
At 20Hz.



b. LDR Sensing and Seven segment display

//IMAGE OF THE CHANGE IN LIGHT INTENSITY FOR DIFFERENT INPUTS (2-3 IMAGES WHICH SHOWS THAT WE CAN SHOW DIFFERENT NUMBERS)





c. Other Modules

(The simulation results of stability feature and average light intensity module is clearly shown in the video recording uploaded.)

5. Key Design Decisions and assumptions

a. Key Design Decisions

- Using a second order Sallen-key filter after calculations
- Making both stability and average filters in the digital domain for better accuracy and more reliable results
- Programmable ICs were not used according to the guidelines

b. Assumptions

- Assumed that the natural ambient light variations are at low frequencies below 10hz.
- Assumed the most interference is from 50-100Hz range
- Assumed stable behavior across small temperature variations in the LDR resistor

6. Conclusion

The Advanced Light Intensity Indicator (ALII) successfully satisfies all of the projects functional criteria by combining digital logic with a carefully created analog filter. The LDRs signals 50-100Hz noise is successfully taken out by the 2nd order Sallen-key low pass filter, making sure that the digital circuit only gets real changes in ambient light. For accurate light level sensing, this gives a solid basis.

The systems give a clear real time display of light intensity from 0-7 using comparators, BDC conversion, and a gate based seven segment decoder. Initially examined using analog techniques, stability mechanism was decided to be fully implemented in the digital domain due to practical limitations. This digital method, which used counters, flip-flops, and adders, was more accurate and reliable and was compatible with both times 30-300s in stability feature and 300-900s in average feature.

Each of the main subsystems were validated through simulation, and the system as a whole, functions precisely to provide reliable and significant light intensity data. Overall, the project produces a dependable, affordable solution, appropriate for practical lighting applications while highlighting important concepts of signal processing, timing control and hardware based computation.

References

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